

Hero of Alexandria, Metrica i Preamble, 1-9

[translated by Henry Mendell \(Cal. State U., L.A.\)](#)

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The text is based on a single manuscript, Cod. Contantinopolitanus Palat. Vet. 1. The translation is based on the edition by E.M. Bruins (Brill, 1964), abbrev., Br. An earlier edition was Hero Alexandrinus, *Opera* iii, ed. Hermann Schöne (Leipzig: Teubner, 1903), abbrev., Sch. Additionally, I have consulted Wilbur Knorr's notes in his copy of Br., abbrev, Kn.

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This text uses unicode. The following translations have been adopted:

ἐμβαδόν = area

χωρίον = plot

μέθοδος = procedure

Heron mostly uses Egyptian fractions (standard in Greek literature from about 300 BCE). These are so-called unit fractions. I shall use:

$\angle = \frac{1}{2}$

$n' = n'$ th marker (e.g, $4' = \frac{1}{4}$)

A number such as $1 \frac{5}{6}$ might be written: $1 \angle 3' (1 + \frac{1}{2} + \frac{1}{3})$

Operations

multiplication: x times y (this will translate the Greek, $\tau\alpha\ x\ \epsilon\pi\iota\ \tau\alpha\ y$)

subtraction: y is taken away from x

squaring: x times itself ($\tau\alpha\ x\ \epsilon\phi'\ \epsilon\alpha\nu\tau\acute{\alpha}$)

square root: side of square x . The term is geometrical in origin, but is used in a numerical sense and may be used geometrically

Geometrical configurations

square: the square from AB (abbrev: that from AB)

rectangle: the rectangle enclosed by AB , $B\Gamma$ (abbrev: that by AB , $B\Gamma$; abbrev: that by $AB\Gamma$)

angle: the angle enclosed by $AB\Gamma$ (abbrev: that by $AB\Gamma$; but usually, angle $AB\Gamma$, to disambiguate from 'rectangle', where the Greek is clear since for 'rectangle', the definite article (trans. "that" or "what's") is neuter, while for 'angle', it is feminine.

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PREAMBLE (pp. 2-6.7)

The first geometry, as the ancient story teaches us, engaged in measurements and divisions on land, whence it was also called 'geometry (land or earth measure)'.¹ Since the subject matter was useful to people, its sort ($\gamma\acute{\epsilon}\nu\omicron\varsigma$) made much progress, so that the administration of measurement and divisions also proceeded to solid bodies. And since the first theorems conceived ($\epsilon\pi\nu\omicron\eta\theta\acute{\epsilon}\nu\tau\alpha$) were not adequate, they required furthermore extraordinary investigation so that even up to now some of them raise difficulties, although Archimedes and Eudoxus nobly contributed to the subject. For there was no means of producing a demonstration before the contrivance ($\epsilon\pi\acute{\iota}\nu\omicron\iota\alpha$ ²) of Eudoxus through which the cylinder is three-times the cone having the same base and height equal to it (Eucl., *El.* xii 10, cf. Arch., *De sphaer. et. cyl.*, intro., and *Meth.*, intro.), and that circles are to each other as the squares from the diameters are to one another. And, before the understanding of Archimedes, it was unthinkable to contrive ($\epsilon\pi\nu\omicron\eta\sigma\alpha\iota$) why the surface of the sphere is four-times the largest of the circles in it (Arch., *De sphaer. et cyl.* i 33), and that its volume is two third-parts of the cylinder circumscribing it (Arch., *De sphaer. et cyl.* i 34), and any siblings of these. And so, the mentioned subject being required, we consider that it would be well to collect everything useful that has been written up by those before us and everything that we've additionally observed. Let us begin from the measurements of planes, including along with planes also the other surfaces, convex and concave, since every surface is conceived ($\epsilon\pi\nu\omicron\epsilon\acute{\iota}\tau\alpha\iota$) from two extensions. And the comparisons of the mentioned surfaces come-about in relation to some rectilinear and rectangular plot: rectilinear, since the straight-line is also invariable in comparison with other lines, since every straight-line fits on every straight-line, while the others, convex and concave lines, do not all of them fit on all of them. [lacuna? Cf. Sch., Br.] Wherefore with regard to something fixed, I mean the straight line, they would furthermore make the comparison with the right angle.³ For again every right angle fits on every right angle, while the others do not all of them fit on all of them. A cubit area is called when a square plot has each side of one cubit. Similarly, a foot area is called, when a square plot has each side of one foot. Thus the mentioned surfaces have comparisons with the mentioned plots or the parts of them. Again, solid bodies then get their relations to a solid, rectilinear and rectangular plot, everywhere equilateral. And this is a cube having each side of either one cubit or one foot. Or again, with the parts of these. And so, it has been stated the reason why the comparison comes-to-be with the mentioned plots, but let us begin in succession with measurements on surfaces. And so, in order that we don't name in each measurement feet or cubits or the parts of this, we will display the numbers in units, since it is possible to suppose them with respect to whatever measure is intended.⁴

This is a variation on a plausible story. Cf. Herodotus, *Hist.* 2.109.

One might think that Hero's frequent use of 'conceive ($\epsilon\pi\acute{\iota}\nu\omicron\iota\alpha$)', a technical term in Stoicism (attending the mind to an object of thought), reveals him a Stoic. It is better to assume that Hero just chooses a ordinary

terminology of his time.

Hero's reasoning is a little odd (every 30° angle fits on every other) unless one thinks of the species of rectilinear angles as acute, right, obtuse.

Heron's use of units ($\mu\nu\acute{\alpha}\delta\epsilon\varsigma$) as an abstract measure departs from the standard practice in the mensuration literaure (Babylonian, Egyptian, and Greek).

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Prop. 1

[\(diagram 1\)](#)

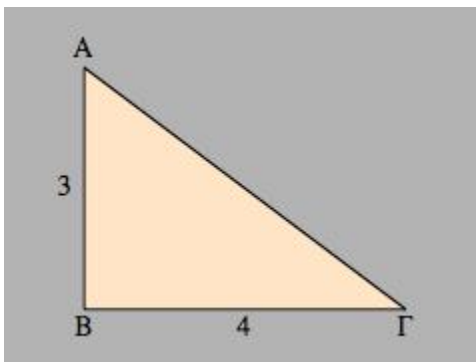


1. Let there be an oblong plot $AB\Gamma\Delta$, having AB as 5 units and $A\Gamma$ as 3 units. To find its area. Since every right-angled parallelogram is said to be enclosed by two straight-lines enclosing a right angle, and what's enclosed by BA , $A\Gamma$ is of this sort, the area of the oblong will be 15 units. [\(diagram 2\)](#) For if each side is divided, AB into 5 units, and $A\Gamma$ similarly into 3 units and through the sections parallels are drawn to the sides of the parallelogram, the plot will be divided into 15 plots, of which each will be 1 unit. And if the plot is a square, the same argument ($\lambda\acute{o}\gamma\omicron\varsigma$) will apply.

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Prop 2

[\(diagram 1\)](#)



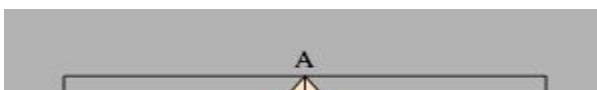
2. Let there be a right-angled triangle, $AB\Gamma$ having as right the angle at B . And let AB be 3 units, and $B\Gamma$ 4 units. To find the area of the triangle and the hypotenuse. [\(diagram 2\)](#) Let right-angled parallelogram $AB\Gamma\Delta$ be filled out, where the area, as was shown above, is 12 units. But triangle $AB\Gamma$ is half of parallelogram $AB\Gamma\Delta$. And so, the area of triangle $AB\Gamma$ will be 6 units; [\(diagram 3\)](#) and since the angle at B is right, the squares from AB , $B\Gamma$ are equal to the square from $A\Gamma$. And the squares from AB , $B\Gamma$ are 25 units. And that from $A\Gamma$ is, therefore, 25 units. $A\Gamma$ itself is, therefore, 5 units.

The procedure is this: making 3 times 4 to take half of these. They become 6. The area of the triangle is so much. ...having made 3 times themselves and similarly 4 times themselves, to add them. And they become 25. And having taken a side of these to have the hypotenuse of the triangle.

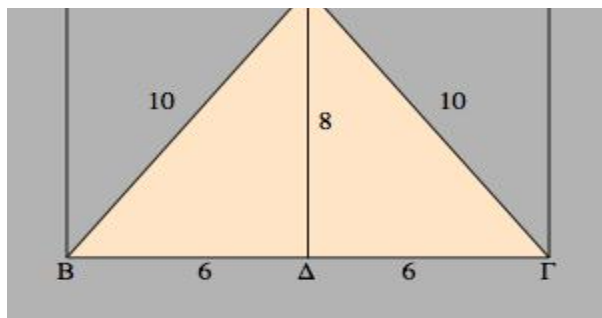
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Prop 3

[\(general diagram = diagram 4\)](#)



[\(diagram 1\)](#) 3. Let there be an isosceles triangle, $AB\Gamma$, having AB equal to $A\Gamma$, and each of the equal units 10, while $B\Gamma$ {to



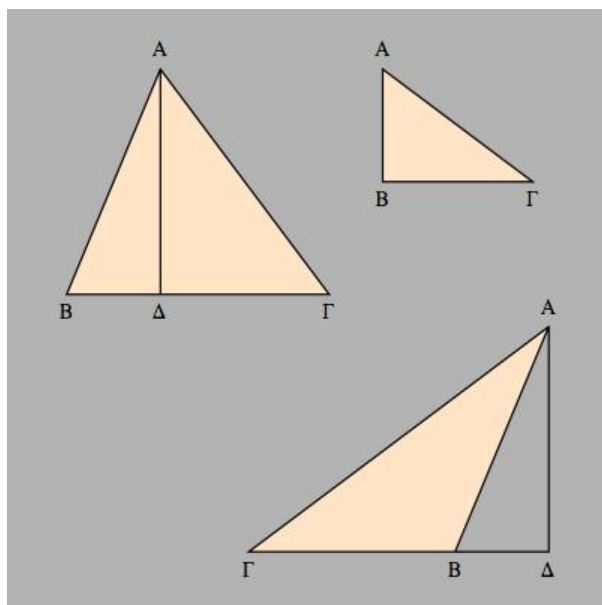
$A\Gamma$ and each of the 10 equal units} $\frac{1}{2}$ of units 12. To find its area. (diagram 2) Let an altitude, $A\Delta$, be drawn to $B\Gamma$. (diagram 3) And through A , let a parallel to $B\Gamma$ be drawn, i.e., EZ , while through B , Γ let parallels to $A\Delta$, be drawn, i.e., BE , ΓZ . Therefore, parallelogram $B\Gamma EZ$ is double triangle $AB\Gamma$. For it has the same base as it and is in the same parallels. And since it is isosceles and $A\Delta$ was drawn perpendicular, $B\Delta$ is equal to $\Delta\Gamma$. And $B\Gamma$ is of 12 units. Therefore, $B\Delta$ is of 6 units. But AB is of 10 units. (diagram 4) Therefore, $A\Delta$ is of 8

units, since the square from AB is equal to those from $B\Delta$, ΔA , so that also BE will be of 8 units. But $B\Gamma$ is of 12 units. Therefore, the area of parallelogram $B\Gamma EZ$ is of 96 units. Thus the area of triangle $AB\Gamma$ is of 48 units. And this is the method. Take half of 12. It becomes 5. And 10 to themselves. They become 100. Take away 6 to themselves, which are 36. They become remainders 64. The side of this becomes 8. Altitude $A\Delta$ will be so much. And 12 against 8. They become 96. The half of these. They become 48. The area of the triangle is of so many units.

1This is clearly a dittography (i.e., where the copyist writes the same phrase twice).

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Prop. 4



Of non-isosceles triangles it is required to observe the angles so that we see the altitudes from the drawn angles to the sides, whether they fall within the angles or outside. And so, let there be a given triangle, $AB\Gamma$, having each side of given portions ($\mu\omicron\rho\omega\delta\nu$). And it is required to observe, as it happens, the angle at A , whether right or obtuse or acute. And so, if the square from $B\Gamma$ is equal to the squares from BA , $A\Gamma$, it is clear that the angle at A is right, while if it is smaller, acute, and if larger, it is clear that the angle at A is obtuse. Let it be supposed, in fact, that the square from $B\Gamma$ is smaller than the squares from BA , $A\Gamma$. Therefore, the angle at A is acute. For if it is not acute, it is either right or obtuse. And so, it is not right. For the square from $B\Gamma$ would have to be equal to the squares from ΓA , AB . But it is not. Therefore, the angle at A is not right. However, it is not obtuse either. For the square from $B\Gamma$ would have to be larger than the squares from ΓA , AB . But it is not. Therefore, it is not obtuse. But it was shown

that it is not right either. Therefore, it is acute. Similarly, in fact, we will also conclude that if the square from $B\Gamma$ is larger than the squares from BA , $A\Gamma$ the angle at A is obtuse.

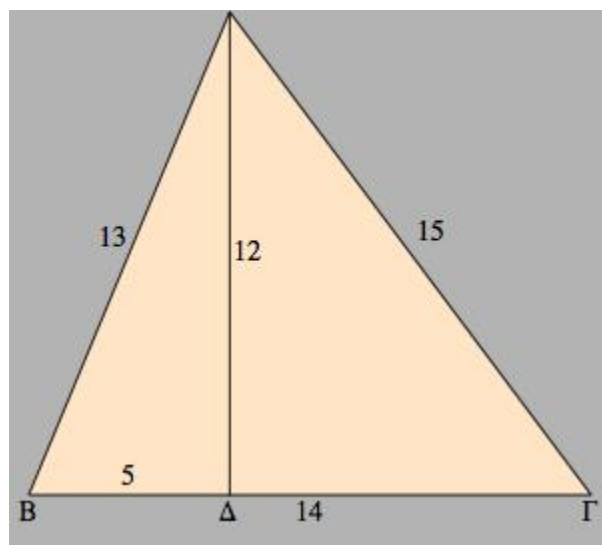
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Prop. 5

(general diagram = diagram 2)



(diagram 1) Let there be an acute-angled triangle, $AB\Gamma$,



having side AB of 13 units, BΓ of 14 units, and AΓ of 15 units. To find its area. It is obvious <lacuna, Sch., Br.> that the angle at B is acute. For the square from AΓ is smaller than the squares from AB, BΔ. (diagram 2) Let a perpendicular, AΔ be drawn to BΓ. Therefore, that from AΓ is smaller than the squares from AB, BΓ by twice the rectangle enclosed by ΓB, BΔ, as has been shown <lacuna, Sch.; in the Elements, Br.> (El. ii 13). And those from AB, BΓ are of 365 units, that from AΓ of 225 units. Therefore, twice the rectangle enclosed by ΓB, BΔ is of 140 units. Therefore, once that by ΓB, BΔ will be of 70 units. And BΓ is of 14 units. Therefore, BΔ will be of 5 units. And since that from AB is equal to those from AΔ, ΔB, and that from AB is of 169 units, while that from BΔ is of 25 units, therefore, a remainder, that from AΔ, will be of 144 units. Therefore, this AΔ will be of 12 units. But BΓ is also of

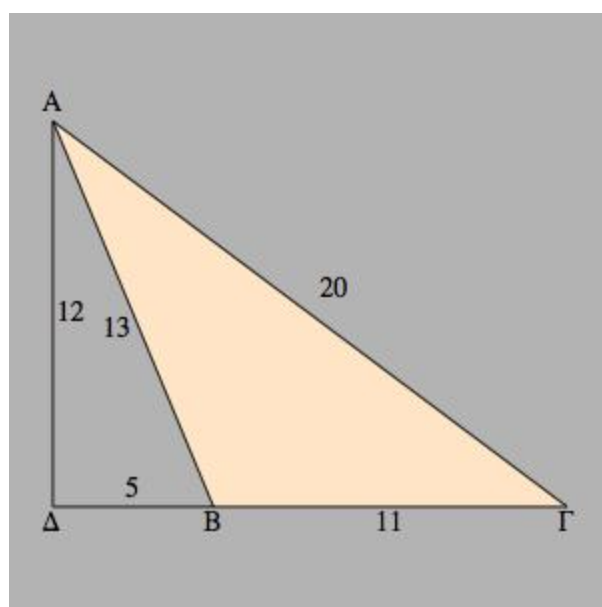
14 units. Therefore, that by BΓ, AΔ will be of 168 units. And it will be double triangle ABΓ. Therefore, triangle ABΓ will be of 84 units.

The procedure will be like this: 13 times themselves. They become 169. And 14 times themselves. they become 196. And 15 times themselves. They become 225. From these take away 224. They become remainders 140. Half of these. They become 70. Apply along 14. They become 5. And 13 times themselves. They become 169. That from which 5 times themselves were taken away. Remainders 144. The side of these become 12. The perpendicular will be of so much. Multiply these times 14. They become 168. Half of these is 84. The area will be of so much.

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Prop. 6

(general diagram = diagram 2)



(diagram 1) 6. Let there be an obtuse-angled triangle, ABΓ having side AB of 13 units, BΓ of 11 units, and AΓ of 20 units. To find its perpendicular and area. (diagram 2) Let BΓ be extended and let an altitude, AΔ, be drawn to it. Therefore, that from AΓ is larger than those from AB, BΓ by twice [the rectangle enclosed] by BΓ, BΔ (El. ii 12). And that from AΓ is of 400 units, while that from BΓ is of 121 units, and that from AB is of 169. Therefore, what's twice by ΓB, BΔ is of 110 units. Therefore, what's once by ΓB, BΔ is of units 55 and BΓ is of units 11. Therefore, BΔ will be of 5 units. But also AB is of 13 units. Therefore, AΔ will be of 12 units. But also BΓ is of 11 units. Therefore, that by AΔ, BΓ will be of 132 units. And it is double triangle ABΓ. Therefore, triangle ABΓ will of 66 units.

The procedure will be this. 13 times themselves become 169. And 11 times themselves become 121. And 20 times

themselves become 400. Add 169 and 121. They become 290. Take these from 400. Remainders 110. Have of these become 55. Apply along 110. They become 5. And 13 times themselves become 169. Take 5 times

themselves. Remainders 144. The side of these become 12. These times 11 become 132. Half of these is 66. The area of the triangle will be of so much.

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Prop. 7

And so, up to this point we have make the geometrical demonstrations by calculating, but in the following we will make the measurements following an analysis through the synthesis of numbers.

7. If there are two numbers AB , $B\Gamma$, the number enclosed by $AB\Gamma$ will be the side of the square from AB times the square from $B\Gamma$. For since as AB is to $B\Gamma$, so is the square from AB to the number contained by $AB\Gamma$ and the [number contained](#) by $AB\Gamma$ to the square from $B\Gamma$, therefore too, as the square from AB will be to the [number contained](#) by $AB\Gamma$, so [will be](#) the [number contained](#) by $AB\Gamma$ to the square from $B\Gamma$. And so since three numbers have proportion, the [number contained](#) by the extremes is equal to the square from the middle. Therefore, the square from AB times the square from $B\Gamma$ is equal to the [number contained](#) by $AB\Gamma$ times itself. Therefore, the number contained by $AB\Gamma$ is the side of the square from AB times the square from $B\Gamma$.

Let the numbers be a, b

$$a \cdot b = \sqrt{a^2 \cdot b^2}$$

$$\text{proof: } a : b = a^2 : a \cdot b \text{ and } a : b = a \cdot b : b^2$$

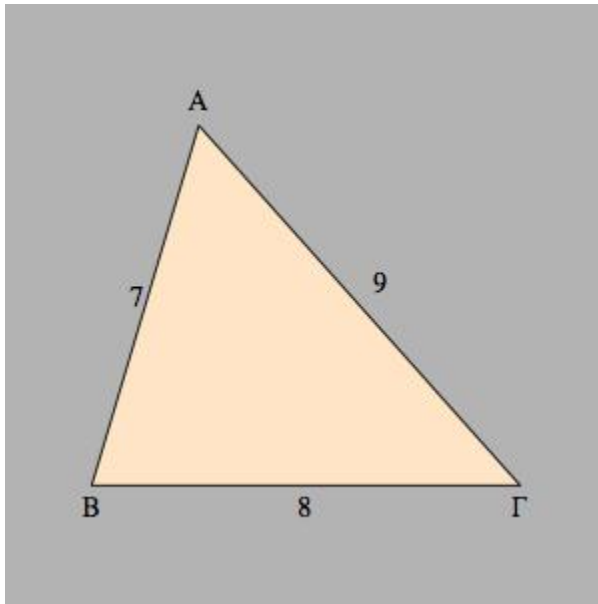
$$\text{therefore, } a^2 : a \cdot b = a \cdot b : b^2$$

$$\text{therefore, } a^2 \cdot b^2 = (a \cdot b) \cdot (a \cdot b) = (a \cdot b)^2$$

$$a \cdot b = \sqrt{a^2 \cdot b^2}$$

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Prop. 8



There is a general procedure for finding the area given three sides of any triangle whatever and without the altitude. For example, let the sides of the triangle be of 7, 8, 9 units. Compose 7 and 8 and 9. They become 24. Take half of these. They become 12. Take away the 7 units. Remainders 5. Again take away the 8 from the 12. Remainders 4. And further the 9. Remainders 3. Make 12 times 5. They become 60. These times 4. They become 240. These times 3. They become 720. Take the side of these and it will be the area of the triangle.

And so, since 720 do not have its side as rational, we will take the side with a least difference in this way. Since a square near to 720 is 729 and has a side 27, partition 720 into 27. They become 26 and two thirds. Add on 27. They become 53 two thirds. Half of these become $26 \frac{1}{3}$. For $26 \frac{1}{3}$ times themselves become $720 \frac{1}{9}$. Thus, the difference is $\frac{1}{9}$ portion of a unit. If we want the difference to come-to-be in a

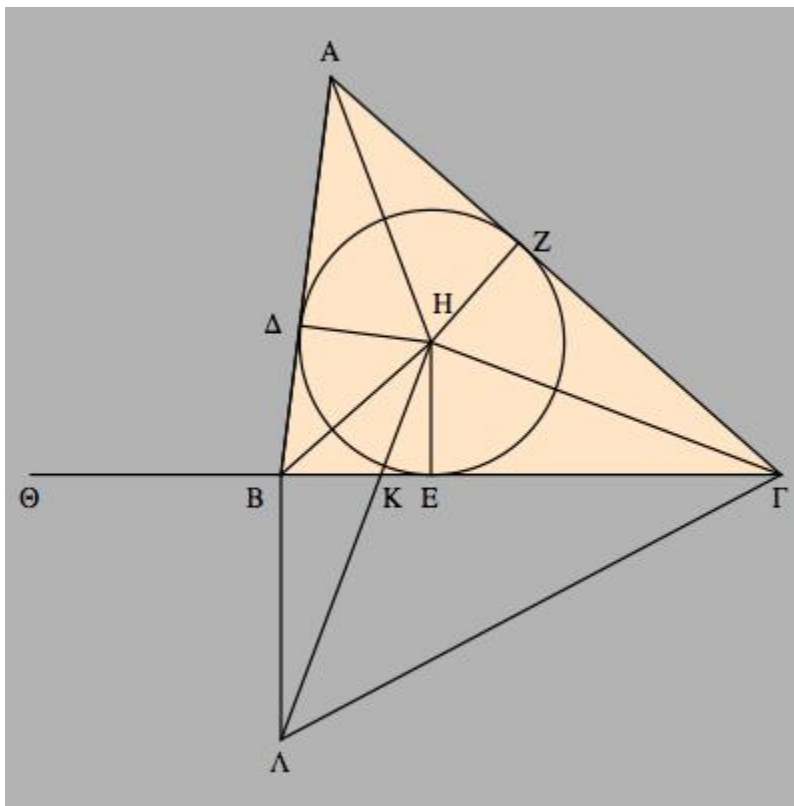
smaller portion than $\frac{1}{9}$, instead of 729 we arrange the presently found 720 and $\frac{1}{9}$, and by doing the same things, we will find the difference becoming much smaller than $\frac{1}{9}$.

Notes: The procedure for the area is: Let the sides of the triangle be a, b, c , and the perimeter $p = a + b + c$. Then, take $p/2 \cdot (p/2 - a) \cdot (p/2 - b) \cdot (p/2 - c)$. Observe that in Greek terms, this is the square of the area. See how this works out in the demonstration. The area of the triangle is then the side of this: $\sqrt{p/2 \cdot (p/2 - a) \cdot (p/2 - b) \cdot (p/2 - c)}$.

Heron also provides a method for approximating square roots. Suppose we want $\sqrt{n}$. Let $\sqrt{p^2}$ be near n . Take $1/2(p+n/p)$ as a first approximation, and proceed if desired. Note that $(1/2(p+n/p))^2 = 1/4(p^2 + 2n + (n/p)^2)$, so that $1/4(p^2 + 2n + (n/p)^2) - n = 1/4(p^2 - 1/2n + (n/p)^2) = 1/4(p - n/p)^2$. To show that the sequence of converge on the original number, one but needs to show the difference between the square of the second approximation and the number is less than half that of the first approximation and the original number (Eucl., *El.* x 1), that if $p^2 > n$, then $p > n/p$ and $1/4(p - n/p)^2 < 1/2(p^2 - n)$, and if $p^2 < n$, then $p < n/p$ and $1/4(n/p - p)^2 < 1/2(n - p^2)$.

Here is the geometrical demonstration of this: the sides of a triangle being given, to find the area. And so, it is possible by drawing a single altitude of it and providing its magnitude to find the area of the triangle, but let it be required to provide the area without the altitude.

([general diagram](#))



([diagram 1](#)) Let the given triangle be $AB\Gamma$, and let each of $AB, B\Gamma, \Gamma A$ be given. To find the area. ([diagram 2](#)) Let a circle, ΔEZ , whose center is H , be inscribed in the triangle (*El.* iv 4), and let $AH, BH, \Gamma H, \Delta H, EH, ZH$ be joined. Therefore, that by $B\Gamma$, EH is double triangle $BH\Gamma$, while that by ΓA , ZH is double triangle $A\Gamma H$, and that by ΓA , ZH is double triangle $A\Gamma H$. Therefore, that by the perimeter of triangle $AB\Gamma$ and EH , that is that from the center of circle ΔEZ , is double triangle $AB\Gamma$. ([diagram 3](#)) Let ΓB be extended, and let $B\Theta$ lie equal to $A\Delta$. Therefore, $\Gamma B\Theta$ is half the perimeter of triangle $AB\Gamma$, due to the fact that $A\Delta$ is equal to AZ and ΔB to B and $Z\Gamma$ to ΓE . Therefore, that by $\Gamma\Theta$, EH is equal to triangle $AB\Gamma$. But that by $\Gamma\Theta$, EH is a side of that from $\Gamma\Theta$ times that from EH . Therefore, the area of triangle $AB\Gamma$ times itself will become equal to that from $\Gamma\Theta$ times that from EH . ([diagram 4](#)) Let $H\Lambda$ be drawn at right angles to ΓH , $B\Lambda$ to ΓB . And let $\Gamma\Lambda$ be joined. And so, since each

of $\Gamma H\Lambda$, $\Gamma B\Lambda$ is right, therefore, quadrilateral $\Gamma H B \Lambda$ is in a circle ([see theorem A below](#)). Therefore, the angles enclosed by $\Gamma H B$, $\Gamma \Lambda B$ are equal to two right angles ([see theorem B below](#)). ([diagram 5](#)) But those by $\Gamma H B$, $A H \Delta$ are also equal to two right angles since the angles at H are bisected by $AH, BH, \Gamma H$ and those by $\Gamma H B$, $A H \Delta$ are equal to those by $A H \Gamma$, $\Delta H B$, and all of them are equal to four right angles ([see theorem C below](#)). ([diagram 6](#)) Therefore, the angle by $A H \Delta$ is equal to the angle by $\Gamma \Lambda B$. ([diagram 7](#)) But a right-angle, by $A \Delta H$, is also equal to a right-angle, by $\Gamma B \Lambda$. Therefore, triangle $A H \Delta$ is similar to triangle $\Gamma B \Lambda$. As, therefore, $B\Gamma$ is to $B\Lambda$, $A\Delta$ is to ΔH , that is $B\Theta$ to EH . And, alternando, as ΓB is to $B\Theta$, $B\Lambda$ is to EH ([diagram 8](#)) that is BK to KE , since $B\Lambda$ is a parallel to EH . And, componendo, as $\Gamma\Theta$ is to $B\Theta$, so is BE to EK . Thus, as that from $\Gamma\Theta$ is to that by $\Gamma\Theta$, ΘB , so too is that by $BE\Gamma$ to that by ΓEK , that is that from EH .

For a perpendicular, EH, has been drawn in a rectangle from the right-angle to the base. Thus that from $\Gamma\Theta$ times that from EH, whose side was the area of triangle $AB\Gamma$, will be equal to that by $\Gamma\Theta B$ times ΓEB . And each of $\Gamma\Theta$, ΘB , BE , ΓE is given. For $\Gamma\Theta$ is half of the perimeter of triangle $AB\Gamma$, while $B\Theta$ is the excess by which half of the perimeter exceeds ΓB , and BE the excess by which half of the perimeter exceeds $A\Gamma$, and $E\Gamma$ the excess by which half of the perimeter exceeds AB , since really $E\Gamma$ is equal to ΓZ , and $B\Theta$ to AZ , since it is also equal to $A\Delta$. Therefore, the area of triangle $AB\Gamma$ is also given.

In the following: $O(x,y)$ is the rectangle formed by lines x and y , $T(x)$ is the square on x , and $S(T(x))$ is the side of the square $T(x)$

Analysis: $H\Delta$ is perpendicular to AB , HE to $B\Gamma$, and HZ to $A\Gamma$, since the sides of the triangle are tangent to the circle. So the radii are altitudes of triangles AHB , $BH\Gamma$, $AH\Gamma$. Hence, triangle $AB\Gamma = BH\Gamma + AH\Gamma + AHB = 1/2O(HE,B\Gamma) + 1/2O(\Delta H,AB) + 1/2O(HZ,A\Gamma) = 1/2O(HE,AB+B\Gamma+\Gamma A) = O(HE, 1/2(AB+B\Gamma+\Gamma A))$. So the difficulty is to find the radius. Since the two tangents from a point outside a circle are equal, $\Delta B = BE$, $E\Gamma = \Gamma Z$, and $A\Delta = AZ$. So $AB + BE + E\Gamma = 1/2(AB+B\Gamma+\Gamma A)$. Since $\Theta B = AB$, $\Theta B + BE + E\Gamma = 1/2(AB+B\Gamma+\Gamma A)$. Hence, $O(HE,\Theta\Gamma) = \text{triangle } AB\Gamma$.

triangle- $AB\Gamma \cdot \text{triangle-}AB\Gamma = O(HE,\Theta\Gamma) \cdot O(HE,\Theta\Gamma) = T(HE) \cdot T(\Theta\Gamma)$. This is not a geometrical operation, but, notoriously, it is a good question what the operation is. In any case, it applies the penultimate step of Prop. 7.

Hero does not actually here draw the inference that triangle- $AB\Gamma = \sqrt{T(HE) \cdot T(\Theta\Gamma)} = O(HE,\Theta\Gamma)$

The trick is to show that $AH\Delta \approx \Gamma B\Delta$. Then, the rest is manipulations of proportions. By the similar triangles, $B\Gamma : B\Delta = A\Delta : \Delta H = B\Theta : EH$ (since, by stipulation, $B\Theta = A\Delta$ and radii $EH = \Delta H$)

$B\Gamma : B\Theta = B\Delta : EH = BK : KE$ (since triangles $EKH \approx BK\Delta$)

$\Theta\Gamma : B\Theta = BE : KE$ (componendo: $\Theta B+B\Gamma : B\Theta = BK+KE : KE$)

$T(\Theta\Gamma) : O(B\Theta,\Theta\Gamma) = O(BE,E\Gamma) : O(\Gamma E,EK)$ ($a : b = c : d \Rightarrow O(a,x) : O(b,x) = O(c,y) : O(d,y)$)

$T(EH) = O(\Gamma E,EK)$ (because right triangles $KEH \approx KH\Gamma \approx HE\Gamma$)

$T(\Theta\Gamma) : O(B\Theta,\Theta\Gamma) = O(BE,E\Gamma) : T(EH)$

$T(\Theta\Gamma) \cdot T(EH) = O(B\Theta,\Theta\Gamma) \cdot O(BE,E\Gamma)$ (the meaning of this is difficult; see above)

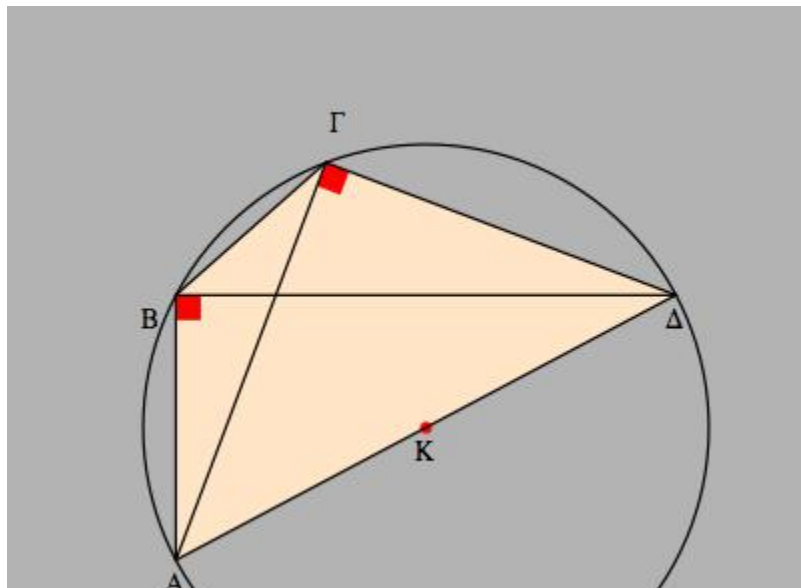
triangle $AB\Gamma = \sqrt{T(HE) \cdot T(\Theta\Gamma)} = \sqrt{O(B\Theta,\Theta\Gamma) \cdot O(BE,E\Gamma)}$

$B\Theta = 1/2 \text{ perimeter} - \Gamma B$

$\Theta\Gamma = 1/2 \text{ perimeter}$

$BE = 1/2 \text{ perimeter} - A\Gamma$

$E\Gamma = 1/2 \text{ perimeter} - AB$



Theorem A: (diagram Thm A) If each diagonal of a quadrilateral makes a right angle with a side, the quadrilateral can be inscribed in a circle,

Proof: Let there be quadrilateral $AB\Gamma\Delta$ and diagonals, $A\Gamma$, $B\Delta$, with angles $B\Gamma\Delta$ and $AB\Gamma$ right. Circumscribe triangle $AB\Gamma$ with circle $AB\Gamma$ (*El. iv 5*). I say that Δ is on the circumference of circle $AB\Gamma E$. Since angle $B\Gamma\Delta$ is right, $A\Delta$ is a diameter (*El. iii 31*). Since angle $AB\Delta$ is also right, it too lies on a semicircle with diameter $A\Delta$, i.e., the same semi-circle.

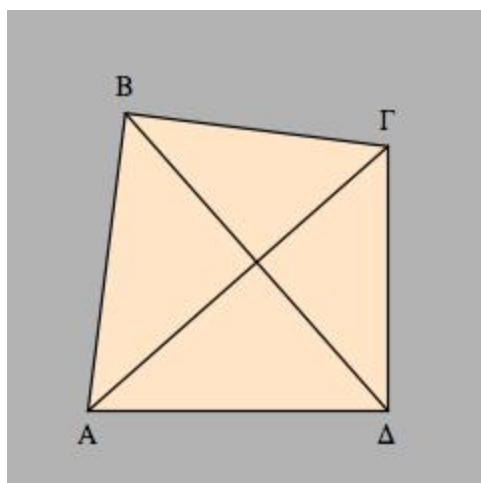


Theorem B: ([diagram Thm B](#)) The opposite angles together of a quadrilateral inscribed in a circle equal two right angles.

Let quadrilateral $AB\Gamma\Delta$ be inscribed in a circle, and diameters $B\Delta$, $A\Gamma$ be joined. I say that $AB\Gamma + \Gamma\Delta\Delta = 2$ right angles. Clearly, angles $BA\Gamma = B\Delta\Gamma$, and $\Delta B\Gamma = \Gamma\Delta\Delta$ (*El. iii 27*). But also angles $BA\Gamma + \Gamma\Delta\Delta + B\Delta\Delta + AB\Delta = B\Delta\Delta + \Delta AB + AB\Delta = 2$ right angles (*El. i 32*). Hence, angles $B\Delta\Gamma + \Delta B\Gamma + B\Delta\Delta + AB\Delta = 2$ right angles, but also $B\Delta\Gamma + B\Delta\Delta = A\Delta\Gamma$, and $\Delta B\Gamma + AB\Delta = AB\Gamma$. Hence, $AB\Gamma + A\Delta\Gamma = 2$ right angles.

Similarly, we could show that $BA\Delta + B\Gamma\Delta = 2$ right angles.

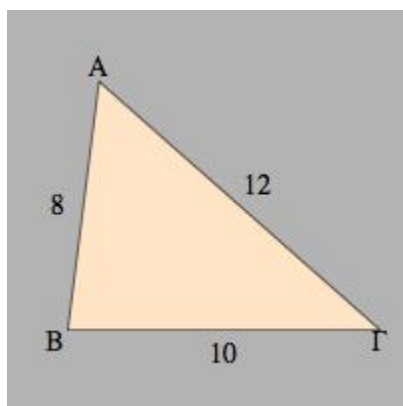
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Theorem C: If a quadrilateral has two adjacent sides equal, and opposite adjacent sides equal, then the line connecting the vertices of the two pairs of adjacent sides bisect the angles.

Let there be a quadrilateral $AB\Gamma\Delta$ with equal sides $AB = B\Gamma$ and $\Gamma\Delta = \Delta A$. Let diameter $B\Delta$ be drawn. I say that $B\Delta$ bisects angle $AB\Gamma$. Let $A\Gamma$ be joined. Since, $B\Gamma = \Gamma\Delta$, $AB = \Delta A$, and $A\Gamma$ is common, triangle $AB\Gamma = ADG$. Therefore, angles $BA\Gamma = B\Gamma\Delta$, and $B\Gamma\Delta = \Delta\Gamma\Delta$.

[return to demonstion](#)



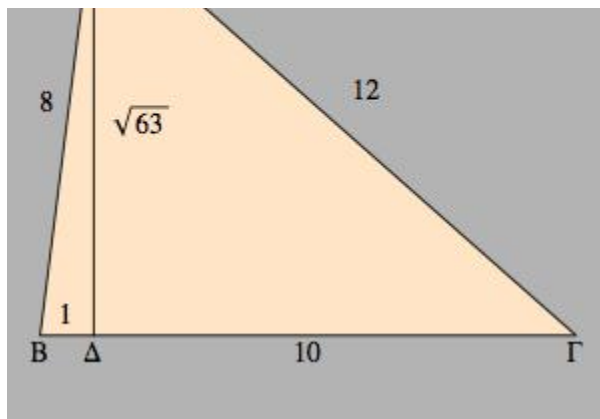
It will be synthesized, in fact, in this way. Let AB be of 13 units, $B\Gamma$ of 14 units, and $A\Gamma$ of 15 units. Add 13 and 14 and 15. And they become 42. half of which become 21. Remove the 13. Remainders 8. Then the 14. Remainders 7. And further the 15. Remainders 6. 21 times 8, and what come-to-be times 7, and further what come-to-be times 6. They are combined 7056. The side of these of so much as 84 will be the area of the triangle.

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Prop. 9



9. And so, after we have learned how, the sides of a triangle being given, to find its area when the altitude is rational, let it be now to find the area when the altitude is not rational. Let there be a triangle $AB\Gamma$, having AB of 8 units, $B\Gamma$ of 10 units, and $A\Gamma$ of 12 units. And let an altitude, $A\Delta$, be drawn.



Following, in fact, what was said in the case of the acute-angles, what's twice by $\Gamma B\Delta$ is of 20 units. Therefore $B\Delta$ will be of 1 unit, and that from it, therefore, of 1 unit. But also that from AB is of 64 units. Therefore a remainder, that from $A\Delta$, will be of 63 units. But also that from $B\Gamma$ is of 100 units. Therefore, that from $B\Gamma$ times that from $A\Delta$ will be of 6300 units. A side of this is the [number] by $B\Gamma$, $A\Delta$ times itself; therefore the [number] by $B\Gamma$, $A\Delta$ times itself will be of 6300 units. Therefore, the half of that by $B\Gamma$, $A\Delta$ times itself will be of 1475. For where the sides of the squares are double one another, those from them are four times those from the halves. But the half of that by $B\Gamma$, $A\Delta$ is the area of the triangle.

Therefore, the area of the triangle is in power 1475. But it is possible by taking the side of 63 [units](#) as near to find the area as if the altitude were rational. But the side $7 \angle 4' 8' 16'$ is near 63. And so it will be required by laying down the altitude of so much to find the area. It is $39 \angle 8' 16'$.

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